Name _____

MATH 146
Introduction to Trigonometry

Fall Semester 2022 (Second Half)

Instructor: Jack B. Smith

EXAM #2, Chapters 8 & 10

Assigned: Tues, Dec 12, 2022
Due: Friday, Dec 16, 2022

INSTRUCTIONS:

- TAKE-HOME EXAM
  - **<u>INDEPENDENT WORK</u> (Honor System)**
  - Do Not Work with others
  - Do Not Seek solutions to these problems at the SMCC Learning Commons
  - **WARNING:** Failure to comply with the honor system will result in a grade of zero and possible suspension from college.
- OPEN BOOK/ OPEN NOTES/OPEN BRIGHTSPACE
- PRINT OUT THE EXAM - ALL WORK IS TO BE PERFORMED IN THIS EXAM BOOK
  - All questions are worth 4 points *unless otherwise noted*
  - Round all angles and magnitudes to **four decimal places** *unless otherwise noted*
  - **CIRCLE ANSWERS** and Include Your Name on first page and initial the following pages
  - **<u>CLEARLY</u> ALL SHOW CALCULATIONS/ <u>NO Credit</u> for Answers Only**
  - **Work must be Neat, Clear and Professional**
- **EXAM SUBMISSION**
  - **Exam must be completed and submitted <u>before midnight</u> on Friday, Dec 16, 2022.**
    - NO EXCEPTIONS – The SMCC 8 Week Semester ends, and letter grades assigned on 12/17/2022.
    - Scan <u>as one complete PDF document</u> and upload your file under Assignments in Brightspace. **Scans must be clean and clear of any grey shading**. Check your scan to ensure that it is clean and clear before uploading.
    - Cell phone photos cannot be accepted.
    - Late submissions cannot be accepted

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BONUS QUESTIONS (1/2 point each):

A) HAVE YOU READ THE EXAM INSTRUCTIONS?

i. Yes _____
ii. No _____

B) Will credit be assigned for illegible work?

i. Yes _____
ii. No _____

C) Are late submissions of this exam permissible?

i. Yes _____
ii. No _____

D) Have you initialed all pages?

i. Yes _____
ii. No _____

E) Is your PDF scan clean and clear of grey shading?

i. Yes _____
ii. No _____

F) If you had a choice to pick MyMathLab (MML cost, $120) or MyOpenMath (MOM cost, Free), which would you pick?

i. MML _____
ii. MOM _____

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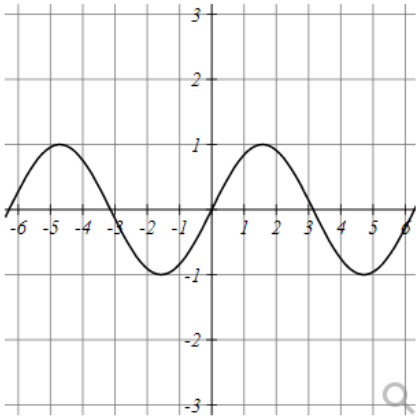
PROBLEM 1.

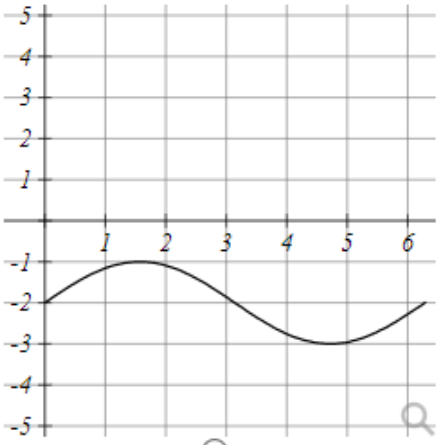


The graph above is NOT a graph of what function?

- ○ y = cos(x)
- ○ y = sin(x)

PROBLEM 2.

Which of the following graphs is the correct plot of $h(x) = \sin(x) + 3$?



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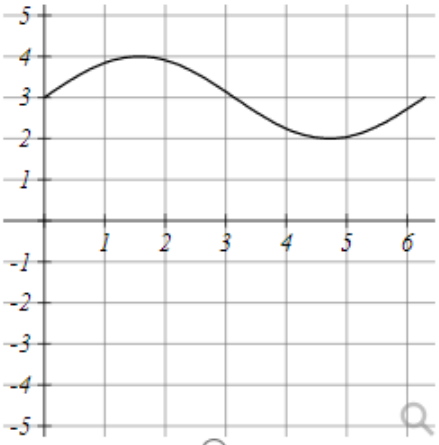

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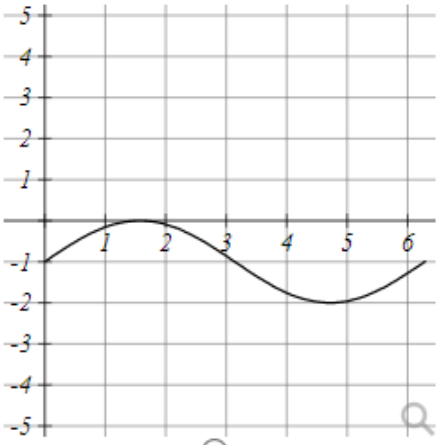

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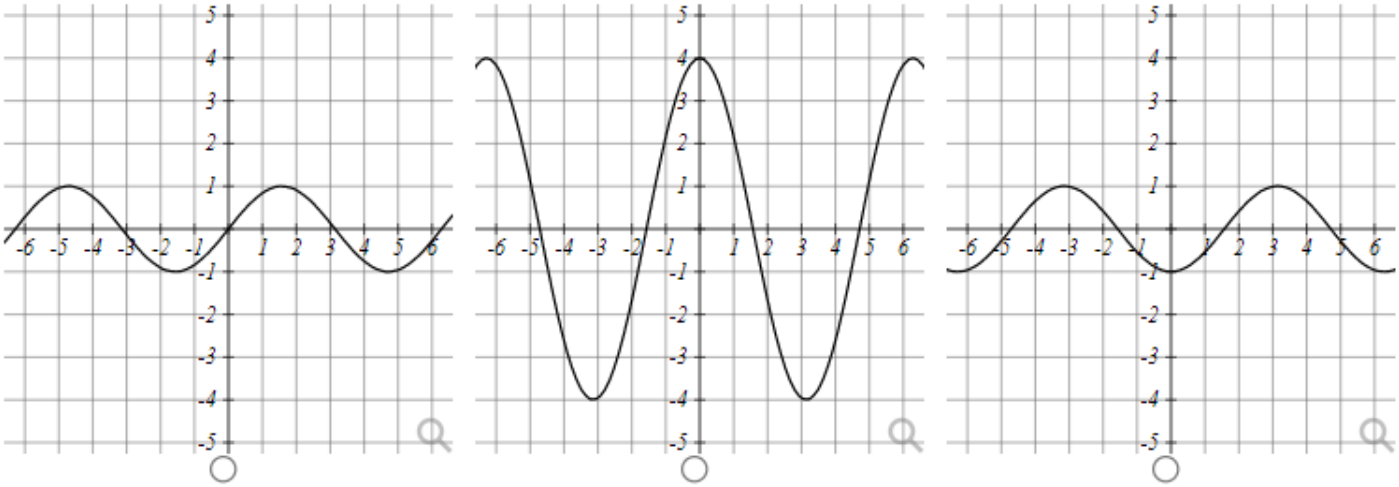
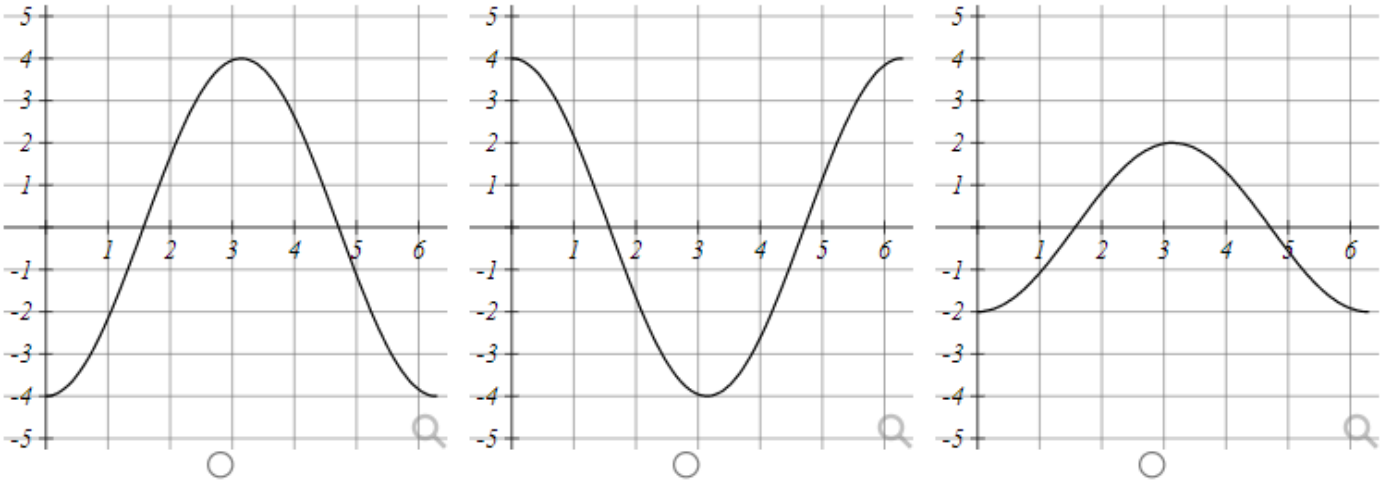
PROBLEM 3.

Which of the following graphs is the correct plot of $y = 4\cos(x)$?



PROBLEM 4.

Which of the following graphs is the correct plot of $y = -4\cos(x)$?



PROBLEM 5.

For: $-7\cos(x) - 4$

Its amplitude = _____

Its midline, y = _____

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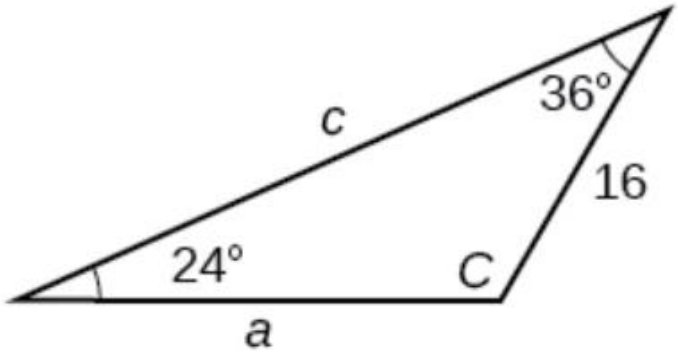
PROBLEM 6.

Assume $\boldsymbol{\alpha}$ is opposite side $\boldsymbol{a}$, $\boldsymbol{\beta}$ is opposite side $\boldsymbol{b}$, and $\boldsymbol{\theta}$ is opposite side $\boldsymbol{c}$. Solve the triangle and *round each answer to the nearest tenth*, given: $\boldsymbol{\beta} = 68°$, $\boldsymbol{b} = 21$, $\boldsymbol{c} = 16$.

(Note: solving a triangle means to find all six components)

PROBLEM 7.

Solve the triangle (*round to the nearest tenth – drawing not to scale*).



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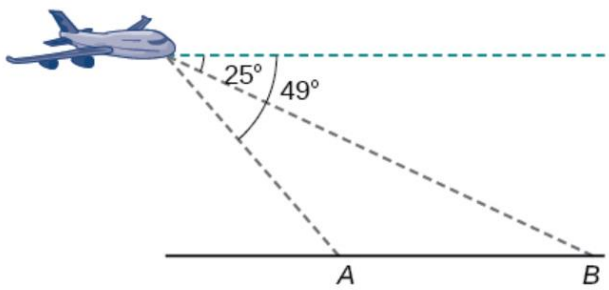
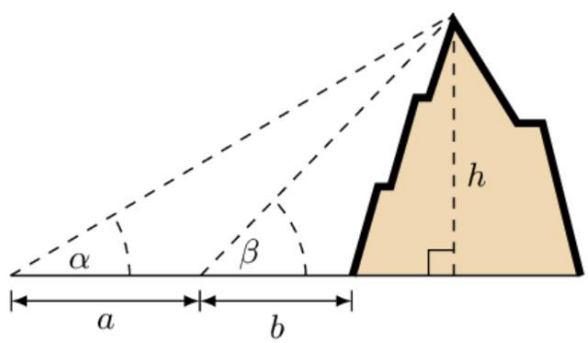
PROBLEM 8.

A pilot is flying over a straight highway. He determines the angles of depression to two mileposts, 2.8 km apart, to be 25° and 49°, as shown below. Find the distance of the plane from point ***A*** and the elevation of the plane (*round your answers to the nearest tenth of a km - illustration not to scale*).



PROBLEM 9.

Find the height of the mountain (see below) if $\boldsymbol{a}$ = 6000', $\boldsymbol{b}$ = 1500', $\boldsymbol{\alpha} = 33°$ *and* $\boldsymbol{\beta} = 39°$ $(round\ your\ answer\ to\ the\ nearest\ foot - drawing\ not\ to\ scale)$.



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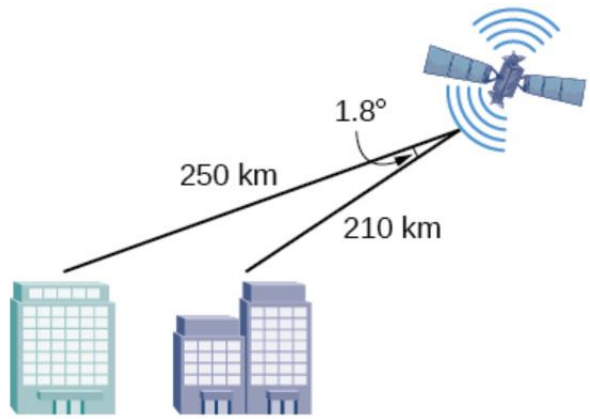
PROBLEM 10.

To find the distance between two cities, a satellite calculates the distances and angle shown below (*not to scale*). Find the distance between the cities. *Round answers to the nearest tenth*.



PROBLEM 11 (6 points).

Solve the triangle, *rounding to the nearest tenth*, assuming $\boldsymbol{\alpha}$ is opposite side $\boldsymbol{a}$, $\boldsymbol{\beta}$ is opposite side $\boldsymbol{b}$, and $\boldsymbol{\theta}$ is opposite side c. Given: $\boldsymbol{a} = 4$, $\boldsymbol{b} = 6$, $\boldsymbol{c} = 8$. *Round answers to the nearest tenth*.

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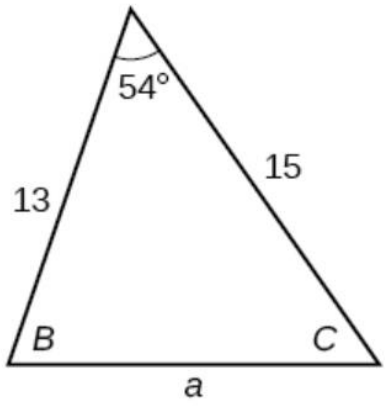
PROBLEM 12 (6 points).

Solve the triangle shown below (*round to the nearest tenth – drawing not to scale*)



PROBLEM 13.

Determine whether the two vectors ***u*** and ***v*** are equal, where ***u*** has an initial point P_1 and a terminal point P_2 and ***v*** has an initial point P_3 and a terminal point P_4.

Given the coordinates: $P_1 = (5, 1)$, $P_2 = (3, -2)$, $P_3 = (-1, 3)$, and $P_4 = (9, -4)$

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PROBLEM 14 (6 points).

Given: $\boldsymbol{u} = \langle 2, -3 \rangle$, $\boldsymbol{v} = \langle 1, 5 \rangle$

Compute:

1. $\boldsymbol{u} + \boldsymbol{v}$
2. $\boldsymbol{u} - \boldsymbol{v}$
3. $2\boldsymbol{u} - 3\boldsymbol{v}$

PROBLEM 15.

Find the magnitude and direction of the vector, $0 \leq \theta < 2\pi$.

Given: $\langle -4, -6 \rangle$

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PROBLEM 16 (9 points).

A 60-pound box is resting on a ramp that is inclined 16°. *Rounding to the nearest tenth*,

a. Make a sketch of this
b. Find the magnitude of the normal (perpendicular) component of the force.
c. Find the magnitude of the component of the force that is parallel to the ramp.

PROBLEM 17.

Given magnitude, $|\boldsymbol{v}| = 6$ and direction, $\boldsymbol{\theta} = 45°$in standard position, write the vector components in the x and y directions.

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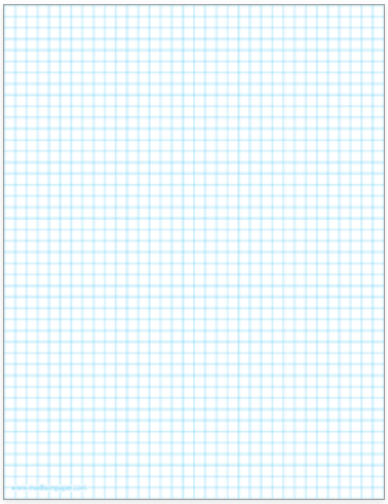
PROBLEM 18.

A woman leaves home and walks 3 miles west, then 2 miles southwest. How far from home is she, and in what direction must she walk to head directly home? *Round answers to two decimal places.*



PROBLEM 19 (8 points).

As part of a video game, a vector from the origin to the point (5, 7) is rotated counterclockwise through an angle of 30° (*Round answers to four decimal places).*

Find the:

1. The final coordinates of the vector.
2. The magnitude of the vector.

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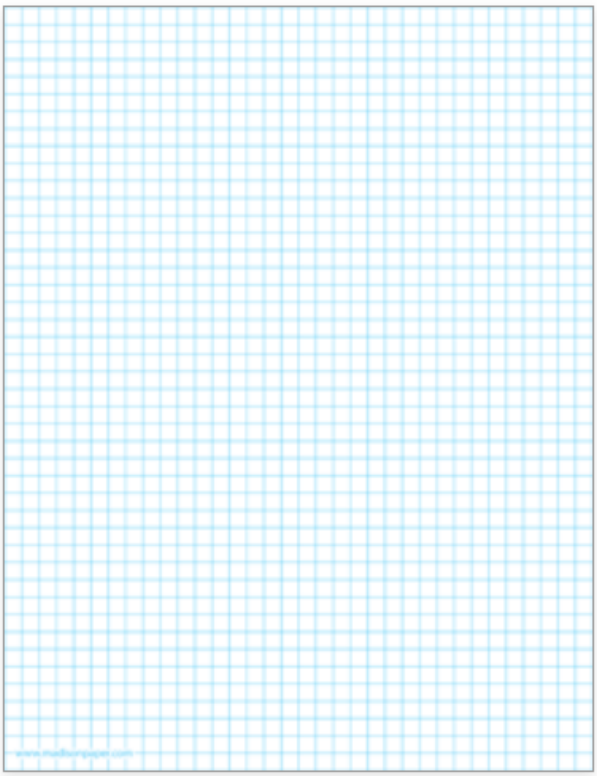
PROBLEM 20 (9 points).

A person starts walking from home and walks:
6 miles East
5 miles Southeast
1 mile West
4 miles South
1 mile East
3 miles Southwest
2 miles East

(*Round answers to two decimal places)*

a. *How many miles has the person walked (1 point)?*
b. *Find the total displacement vector of this walk (4 points)*
c. *If this person walks straight home, how many miles will they walk (4 points)?*



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Graph paper for sketches/calculations/etc